

What City Would You Live In?

We aimed to explore the dynamics of cost of living across several key cities in Asia and the United States. Our focus narrowed to four distinct locations: Tokyo, Hanoi, Bangkok, and Houston. These selections showcase a variety of economic landscapes, from the fast-paced urban life of Tokyo to the rich cultural tapestry of Hanoi. Our goal was to investigate living expenses and how they fluctuate in relation to differing economic environments. We anticipated that costs would align with each city's economic profile, predicting Tokyo would emerge as the most expensive, followed by Bangkok, Houston, and then Hanoi. However, our initial assumptions gave way to unexpected findings as we examined the data more closely.

		Tokyo Japan			Bangkok Thailand			Hanoi Vietnam			Houston Texas			Averages		
Category	Bill	2021	2022	2023	2021	2022	2023	2021	2022	2023	2021	2022	2023	2021	2022	2023
Housing, Utilities, & Transportation	Rent (Single bed apt)	\$ 765.20	\$ 707.03	\$ 664.22	\$301.97	\$291.12	\$334.16	\$ 257.90	\$ 228.92	\$ 244.95	\$ 1,095.47	\$ 1,274.59	\$ 1,175.00	\$ 605.14	\$ 625.42	\$ 605.14
	Public Transportation	\$ 20.70	\$ 15.10	\$ 13.00	\$ 13.90	\$ 12.20	\$ 12.40	\$ 3.00	\$ 3.30	\$ 3.30	\$ 12.00	\$ 17.50	\$ 12.50	\$ 12.40	\$ 12.03	\$ 12.40
	Utilities (Elec, heat, Water)	\$ 201.37	\$ 178.45	\$ 220.58	\$ 80.94	\$ 80.58	\$ 93.99	\$ 56.17	\$ 93.99	\$ 83.46	\$ 143.97	\$ 170.48	\$ 199.54	\$ 120.61	\$ 130.88	\$ 130.88
	Internet bill	\$ 40.42	\$ 36.04	\$ 35.44	\$ 18.59	\$ 15.84	\$ 16.56	\$ 10.11	\$ 8.73	\$ 9.49	\$ 58.63	\$ 73.91	\$ 65.16	\$ 31.94	\$ 33.63	\$ 33.63
Food and Grocery	Loaf of bread	\$ 2.40	\$ 1.44	\$ 2.02	\$ 1.47	\$ 1.89	\$ 1.56	\$ 1.02	\$ 1.08	\$ 1.25	\$ 2.50	\$ 3.12	\$ 3.17	\$ 1.85	\$ 1.88	\$ 1.88
	Milk 1L	\$ 1.84	\$ 1.50	\$ 1.59	\$ 1.78	\$ 1.67	\$ 1.75	\$ 1.49	\$ 1.39	\$ 1.58	\$ 0.80	\$ 0.95	\$ 0.98	\$ 1.48	\$ 1.38	\$ 1.38
	Fast food meal	\$ 9.07	\$ 8.19	\$ 8.25	\$ 2.52	\$ 2.66	\$ 3.62	\$ 2.13	\$ 2.91	\$ 2.63	\$ 17.84	\$ 18.02	\$ 17.77	\$ 7.89	\$ 7.95	\$ 7.95
	Water 1.5L	\$ 1.29	\$ 1.01	\$ 0.96	\$ 0.56	\$ 0.48	\$ 0.45	\$ 0.55	\$ 0.75	\$ 0.53	\$ 1.62	\$ 2.33	\$ 1.91	\$ 1.01	\$ 1.14	\$ 1.14
Lifestyle	Gym Membership	\$ 93.56	\$ 76.23	\$ 65.87	\$ 63.73	\$ 55.39	\$ 62.27	\$ 26.46	\$ 27.93	\$ 25.69	\$ 43.50	\$ 46.32	\$ 51.08	\$ 56.81	\$ 51.47	\$ 51.47
	Pair of Jeans	\$ 61.83	\$ 48.78	\$ 59.70	\$ 64.40	\$ 55.09	\$ 56.94	\$ 28.26	\$ 43.45	\$ 34.78	\$ 47.56	\$ 47.11	\$ 49.09	\$ 50.51	\$ 48.61	\$ 48.61
	Nike Shoes	\$ 89.26	\$ 63.00	\$ 68.99	\$ 91.08	\$ 96.30	\$107.22	\$ 77.87	\$ 97.67	\$ 93.71	\$ 69.31	\$ 87.21	\$ 107.18	\$ 81.88	\$ 86.05	\$ 86.05
	Movie Ticket	\$ 17.35	\$ 14.14	\$ 13.23	\$ 7.49	\$ 7.89	\$ 7.73	\$ 4.19	\$ 4.76	\$ 4.50	\$ 10.84	\$ 13.53	\$ 14.91	\$ 9.97	\$ 10.08	\$ 10.08
Total		\$ 1,304.29	\$ 1,150.91	\$ 1,153.85	\$648.43	\$621.11	\$698.65	\$ 469.15	\$ 514.88	\$ 505.87	\$ 1,504.04	\$ 1,755.07	\$ 1,698.29	\$ 981.48	\$ 1,010.49	\$ 1,010.49
Average				\$1,203.02			\$656.06			\$496.63			\$1,652.47			\$1,010.49
	lifestyle averages		\$ 51.95			\$ 58.54			\$ 39.67			\$ 55.57				
	food averages		\$ 3.30			\$ 1.70			\$ 1.44			\$ 5.92				
	Internet Average		\$ 37.30			\$ 17.00			\$ 9.44			\$ 65.90				

Sources:

Tokyo cost

Bangkok cost

Hanoi cost

Houston cost

Our analysis covers a three-year period (2021-2023), detailing various living expenses such as Housing, Utilities & Transportation, Food & Grocery, and Lifestyle.

The table highlights clear trends in living expenses, with notable differences across the cities. For example, Houston shows the highest living costs overall, driven by high rent and utility prices. On the other hand, Hanoi remains the most affordable, with consistent decreases in living expenses across multiple categories. The color-coding and sparklines visually emphasize the trends, with general price increases across most cities from 2021 to 2023.

Despite some expectations, Tokyo's expenses declined in several categories by 2023, notably in rent and public transportation costs. Bangkok shows steady increases in fast food and lifestyle costs but remains comparatively affordable. These findings

challenged our assumptions, particularly in the case of Houston, which had higher-than-expected costs when compared to other cities. This suggests that local economic conditions, rather than simple national trends, may better explain cost variations across regions.



Our analysis included a bar chart, highlighting the comparative costs of living in four cities. While Houston emerged as the most expensive, costing \$1,698.29, Tokyo followed closely at \$1,153.85. Bangkok's living expenses were significantly lower at \$988.65, and Hanoi stood out with the most affordable cost of living at \$505.87. Houston's high costs could be attributed to its economic strength as an energy hub, whereas Tokyo's pricing reflects its role as a global financial center. Meanwhile, the relatively lower costs in Bangkok and Hanoi highlight more affordable living conditions in Southeast Asia.

Looking at the various statistics and charts to compare, it is obvious that some locations have an advantage in one factor over the others, so when it comes to ranking it is a matter of how much value that advantage brings. For example, Bangkok is by far the least expensive in terms of rent, however the cost of lifestyle decisions such as a gym membership is more expensive than in Tokyo. That said, the price difference of a gym membership in Tokyo vs Bangkok would be around a \$5-\$20 difference while the average rent cost is a steep \$400 difference. Due to the overall value of the cheaper prices the best ranking for these locations is as follows

1. Bangkok: The overall cheaper costs for everything in relation to the other cities
2. Hanoi: for the same reason as Bangkok, however still slightly more costly

3. Tokyo: costs are cheaper than Houston but still significantly more than Bangkok or Hanoi
4. Houston: Inarguably the most expensive city in this list by far.

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Category	Tokyo	Bangkok	Hanoi	Houston	Category	Tokyo	Bangkok	Hanoi	Houston	Category	Tokyo	Bangkok	Hanoi	Houston	Category	Tokyo	Bangkok	Hanoi	Houston
Overall Price	3	2	1		Price	3	2	1		Price	4	2	1		Price	3	4	3	3
Lifestyle	2	3	1		Entertainment	3	2	4		Entertainment	3	2	1		Entertainment	4	3	2	3
Internet	3	2	1		Internet	3	2	1		Internet	3	2	1		Internet	3	2	1	4
Food	3	2	1		Food	3	2	1		Food	3	2	1		Food	2	4	2	4
Average	2.75	2.25	1		Average	3	2	1.75	3.25	Average	3.25	2	1	3.75	Average	3	3.25	2	3.5
Min Value	1				Min Value	1.75				Min Value	1				Min Value	2			
Max Value	4				Max Value	3.25				Max Value	3.75				Max Value	3.5			
Total factors involved	4				Total factors involved	4				Total factors involved	4				Total factors involved	4			

We present charts that assess our individual preferences and rankings for each city across various categories we felt appropriate. While some locations might be better than others in safety, economically, politically, whatever the issues may be. When it came to these four factors Hanoi would rank the best even though it was not the cheapest. And Houston would rank the ‘worst’ due to its expensive lifestyle. Tokyo would be a close third. While Tokyo is a genuinely nice place to live, it is also extremely expensive. And Bangkok would rank 2nd. It would be a little bit more expensive than Hanoi.

In conclusion, our research surprised us. With us living close to Texas we would understand how expensive Houston would be, but did not realize how expensive it really would be. It was also surprising to see how cheap places like Bangkok and Hanoi are. While Tokyo is one of the nicest cities, it did not surprise us that it was as expensive as it was.

Task Tracking

Manage Worksheets and Workbooks

1. Changed height of Cells E through S.
2. Changed width of Column K with Autofit.
3. Changed width of Column A to an approximate value 4. Changed width of Columns to an approximate value.
5. Inserted hyperlink in text form in Cell A7.
6. Edited Document Properties. Title “Group 6 Project”.
7. Used Document Inspector to remove header and footer and check for logical issues within spreadsheet. Fixed errors with Document Inspector

Manage Data Cells and Ranges

1. Middle aligned all cells with header titles
2. Center aligned all cells with header titles
3. Bold font for City titles
4. Rotated cell contents

5. Used Format Painter tool to make city titles uniform versus manually inputting the changes (font, rotation, size, etc.)
6. Added "All Borders" style to Cells C7-D21.
7. Added top and thick bottom border style to Cells Q7-S21.
8. Applied Currency format to Cells.
9. Wrapped text in Cell D8
10. Changed cell fill color in Cell L24.
11. Added Conditional Formatting Rule. Highlight Cell red if greater than Cell S21.
Added Conditional Formatting Rule. Highlight Cell green if less than Cell S21.
12. Inserted sparkline graph in cells F21, I21, L21, O21, R21

Manage Tables and Table Data

1. Applied different table styles to all 6 tables for clarity.
2. Applied Total Row to Tables.
3. Filter buttons are available on 3 main tables to filter results.

Perform Operations by Using Formula and Functions

1. Total Row all utilize SUM function
2. AVERAGE function utilized in cells G21, J21, M21, P21, S21
3. SUM function utilized in cells E20:S20
4. Max functions are utilized on bottom of Preference tables to show difference between all users.
5. Min functions are utilized on bottom of Preference tables to show difference between all users.
6. Used COUNT function in preference tables to show 'total factors involved'

Manage Charts

1. Inserted Clustered Column chart
2. Changed chart title
3. Added Y-axis label "Cost (Dollars)"
4. Added data labels to chart elements. Placed them on "Inside End" and changed font color.
5. Changed Axis Elements Major amount to "1000.0" for clarity on Y-Axis.
6. Used "Select Data" in Chart Elements to change the text in legend from Series 1-4